

Harsha Vardhan Reddy

Software Engineer | B.Tech CSE Graduate (2026) | Open to All Locations

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PROFESSIONAL SUMMARY

B.Tech CSE graduate (2026) with hands-on experience in **software development, backend engineering, data processing, quality assurance, and process automation**. Proficient in **Java and Python** with a strong foundation in OOP, data structures, algorithms, design patterns, and the full SDLC. Practical experience building scalable REST APIs, agentic AI pipelines, cloud-deployed applications, and automated test suites. Holds certifications in **Generative AI (Google Cloud)** and **RPA Development (Automation Anywhere)**. Strong analytical thinker with excellent written and verbal English communication, a collaborative mindset, and a passion for building reliable, impactful technology solutions.

WORK EXPERIENCE

Microsoft & Edunet Foundation

May 2025 – Jun 2025

Application Development Intern | Azure Cloud Platform | Remote

- Designed and developed **3 end-to-end application workflows** on Microsoft Azure following Agile and SDLC best practices, reducing manual processing time by 40% and improving throughput by 25%.
- Performed **root-cause analysis and systematic debugging** across 5 distributed data pipelines, resolving critical defects and improving system reliability by 30% with failure rate below 2%.
- Designed and executed **integration, regression, and functional test cases** validating all application modules under normal, boundary, and failure conditions — achieving zero critical defects before deployment.
- Collaborated with a **cross-functional Agile team of 10+**; participated in code reviews, sprint planning, and authored technical documentation adopted as the team's official reference guide.

EDUCATION

Vignan's Foundation for Science, Technology & Research, Guntur

Aug 2022 – May 2026

B.Tech – Computer Science and Engineering | CGPA: 7.0 / 10.0 | No Active Backlogs

Tirumala Junior College, Rajahmundry

Jun 2020 – Apr 2022

Intermediate – MPC | 63.2%

Tirumala E.M School

Jun 2019 – Apr 2020

SSC | CGPA: 10.0 / 10.0

PROJECTS

AI-Powered IT Service Desk Automation Agent

2024 – 2025

Java, Python, FastAPI, LangGraph, MongoDB Atlas, JWT, OAuth2, bcrypt | Scalable agentic backend system

- Architected a **5-node agentic pipeline** using Python and FastAPI with full async/await processing, automating IT service desk ticket classification and routing — reducing manual effort by 70%.
- Built **9 RESTful API endpoints** applying OOP, data structures, and algorithm design; implemented JWT, OAuth2, and RBAC for secure, stateless authentication and credential management.
- Designed **MongoDB Atlas data schema** with full CRUD operations, enabling 100% cross-session data consistency and reliable incident history tracking across distributed nodes.
- Conducted **end-to-end unit, integration, and regression testing** across all modules; maintained defect logs and technical documentation throughout the full SDLC — achieving zero critical production defects.

AI Chatbot for IT Support Query Resolution

2024

Java, Python, NLP, TF-IDF, Random Forest, Scikit-learn | Intelligent ML classification system

- Built an **NLP-based classification system** resolving 100+ IT query types across 15+ intent categories with **97% accuracy and 0.97 F1-score** using TF-IDF and Random Forest algorithms.
- Optimized the **data processing and inference pipeline** through algorithmic tuning, achieving 50% reduction in latency and validating stability via regression and scale testing.

SKILLS

Programming	Java, Python OOP, Data Structures, Algorithms, Design Patterns, Multithreading
Backend & APIs	FastAPI, REST API Design, Async Programming (async/await), Microservices (Familiar)
AI / ML	LangGraph, LangChain, Agentic AI, NLP, TF-IDF, Random Forest, Scikit-learn, LLMs
Security	JWT, OAuth2, RBAC, bcrypt, Secure Credential Management
Databases	MongoDB Atlas (NoSQL), MySQL, PostgreSQL (SQL – Familiar)
Cloud & DevOps	Microsoft Azure, Azure Cognitive Services, Git, GitHub, Docker (Familiar), CI/CD (Familiar)
Testing & QA	Unit Testing, Integration Testing, Regression Testing, Test Case Design, Root-Cause Analysis
Data Processing	Data Pipeline Design, ETL Concepts, Process Automation, Data Validation, Classification
Web	HTML, CSS, JavaScript (Basic) Responsive Web Design
Process & Methods	Agile (Scrum), SDLC, TDD, Code Reviews, Technical Documentation, Knowledge Sharing
Soft Skills	Analytical Thinking, Problem Solving, Communication, Team Collaboration, Adaptability

CERTIFICATIONS

- Generative AI Fundamentals** – Google Cloud Skills Boost (2024) | Vertex AI, PaLM 2, LLM deployment, Prompt Engineering
- Certified RPA Developer** – Automation Anywhere University (2024) | Enterprise RPA bot development, Business Process Automation, Bot Deployment

ACHIEVEMENT

- Technical Event Coordinator** – Vignan Mahotsav 2025 | Led a 15-member team across 4 technical competitions with 200+ participants; demonstrated leadership, strategic planning, and cross-functional coordination achieving 95% participant satisfaction rate.